Lab 2: Speaker Characterization

Objective

Determine the physical characteristics of a speaker and speaker box. This lab is based upon a Freshmen Year Experience course at Penn State University.

Equipment List and Setup

Item	Quantity
speaker	1
speaker box	1
frequency generator	1
multimeter & probes	1
100Ω resistor	1
digital oscilloscope	1
electronic switching box	1
5 gram masses	5

Theory

A speaker is a beautiful electromechanical system. Magnetic force is exerted when AC current flows through a coil of wire in a magnetic field. The coil is attached to a paper cone which is connected to a flexible surround and finally to the speaker frame. The flexible surround has a spring constant and the wire coil plus cone has a mass so you can imagine there will be a resonance frequency for the system in addition to mechanical damping. The first part of this lab will be devoted to finding this resonance frequency and the second to determining the mechanical damping.

We don't want to destroy the speaker to determine the mass of the coil, the suspension's spring constant or the mechanical resistance of the system. These values are important for modeling the speaker's performance in a speaker box (which has its own acoustic properties!), experiment or audio environment. Additionally, the measurement of these values on a production line gives a means of on-the-spot quality control and testing. The method we will employ in this lab is generally called non-destructive testing.

Effective Mass and Stiffness:

We can be clever and note that for a simple mass and spring we have the relationship,

$$f_0 = \frac{1}{2\pi}\sqrt{\frac{s}{m_0}}, \tag{1}$$

where s is the effective spring constant of the speaker and m_0 is its effective mass. We can easily change the effective mass of the system by adding mass to the dome of the speaker. This would result in a new resonance frequency. Hence,

$$f_i = \frac{1}{2\pi}\sqrt{\frac{s}{m_0 + m_i}}. \tag{2}$$

This can be rearranged and solved for the period-squared,

$$\frac{T_i^2}{4\pi^2} = \frac{m_0}{s} + \frac{m_i}{s}, \tag{3}$$

which looks like the equation for a straight line with slope equal to $1/s$ and y-intercept equal to m_0/s. Thus, by determining the resonance frequencies for the system with various added masses we can determine the unknown effective mass, m_0, and the effective stiffness (spring constant), s.

Mechanical Resistance:

To determine the mechanical resistance of the speaker we need to measure its decaying oscillations as a function of time. To do so we remove all masses from the speaker dome and retune the drive frequency to the resonance frequency. To measure the free decay oscillations we'll make use of a switch box. When the switch is thrown it will disconnect the signal generator from the speaker and tell the digital o-scope to capture the resulting damped oscillations. These oscillations are measured by the oscilloscope hooked up to the speaker's voice coil. As the voice coil moves in the gap between the poles of the speaker's magnet a voltage is induced due to Faraday's Law. The oscilloscope records this voltage on its screen. Note that we are measuring the mechanical resistance, not mechanical *and* electrical resistance!

Recall that the exponential decay of the **velocity amplitude**[1] for the system is given by,

$$u_{max}(t) = u_0 e^{-\beta t}, \tag{4}$$

where $\beta = R_m/2m_0 = 1/\tau$ with R_m as the mechanical resistance, m_0 is the speaker's effective moving mass and τ is the *time constant* which is the time for the amplitude to decay to $1/e$ of its initial value. However, the voltage we measure is proportional to the velocity of the coil. Luckily, we are only interested in the time constant, not the leading constant which would depend on the magnetic field strength and the number of coils (ie, the Bl factor!). Hence, we can write for the decaying voltage signal,

$$V_{max}(t) = V_0 e^{-\beta t}. \tag{5}$$

We will use the cursors on the digital oscilloscope screen to locate the voltage peak and trough values and their corresponding times. Lets label the peak and troughs with the index, n. Now, we can write Eq 6 as,

$$V_n(t) = V_0 e^{-\beta t_n}. \tag{6}$$

[1]Note: We could record and *fit* the entire damped sinusoidal waveform but this is not necessary.

n	t_n	V_n	$\ln(V_n)$
0.0			
0.5			
1.0			
1.5			
$\vdots$			

Table 1: Free Decay Data Table

Lastly, we can take the natural log of both sides to make a linear function which yields,

$$\ln(|V_n|) = \ln(V_0) - \beta t_n. \tag{7}$$

Table 1 illustrates the data to be recorded. The slope of the best fit line through t_n vs $\ln(|V_n|)$ equals β which is related to R_m via $\beta = R_m/2m_0$.

Procedure

We will test a 10-inch woofer and its closed speaker box. Be careful to always use a current limiting resistor (100Ω) with the speaker during the resonance frequency tests to avoid damaging the voice coil. Remove the resistor when doing the free decay experiment.

Effective Mass and Stiffness:

Turn on the oscilloscope and signal generator. Set the signal generator to give a peak-to-peak voltage of 10V. Use the BNC-to-Alligator Clip cables to connect the signal generator, speaker, current limiting resistor and oscilloscope as illustrated in Fig. 1.

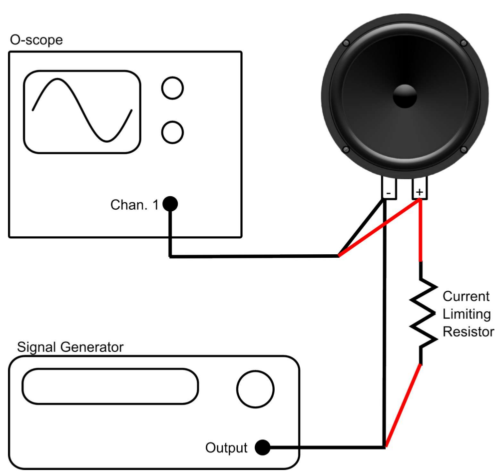

Figure 1: Setup for determining the effective mass and stiffness of the woofer.

1. Start with the frequency generator set to 30Hz. Turn the knob on the signal generator to change the frequency. The O-scope should be displaying a sine wave that represents the voltage across the speaker voice coil.

2. Increase the frequency by turning the knob. You should see the wave on the screen increase in amplitude. Continue adjusting the frequency until the amplitude is at a maximum. You may need to adjust the horizontal and vertical scaling.

3. Record this frequency and the amount of added mass (0g in this case) in a data table.

4. Decrease the frequency and add 5g to the center of the speaker's dome. Again find the resonance frequency by observing the speaker's response on the screen.

5. Repeat this process until you have added 30g to the speaker dome.

When you are done with the data collection plot and analyze the data as suggested above in the theory section. Determine the effective mass, m_0, and the effective stiffness, s, of this speaker. Now, place the speaker into the speaker box (minus the foam) making sure you have connected the electrical wiring correctly and securely screw the speaker into position. Repeat the above procedure with the added mass to again determine the effective mass and effective stiffness. What does the speaker box do, if anything, to these quantities? Why do you think this happens? Do it again with fluff in the box!

Mechanical Resistance:

Remove the speaker from the speaker box and set the function generator to the m_0 resonance frequency (that is, the resonance frequency with zero added mass) and 3 V_{pp}. Use the BNC-to-BNC cables to make the following connections between the speaker, signal generator, o-scope, and switch box.

Switch Box Connections:

1. Connect the "Scope" BNC to the oscilloscope "Ch. 1" input.

2. Connect the "Trig." BNC to the oscilloscope's "External Trigger" input.

3. Connect the voice coil of the speaker to the "Coil" BNC.

4. Connect the "Gen." BNC to the signal output of the function generator.

5. Set the switch to the "Drive" position. (Green LED should be on.)

External Trigger O-scope:

To acquire a single free-decay waveform, the o-scope has to trigger on the signal from the switch box (there is a 9V DC battery inside).

1. Press the "Mode/Coupling" button, then use the "soft keys" to the right of the display to choose the following options. Mode→Sweep, Coupling→DC, triggering source→External.

2. Use the "Horizontal" knob to increase the sweep time to 20 msec/div so that more of the exponentially decaying waveform will me displayed.

When the above preparations are complete you are ready to acquire data. Flip the switch and observe the screen. You should observe a decaying sinusoidal wave. Flip the switch gently a few times waiting a second or so between flips until you have a quality graph with many oscillations. You may need to adjust the horizontal and vertical scale to observe more of the wave on the screen.

Once you have a nice decaying sinusoidal graph press the "Cursor" key and use the soft keys to the right of the display to set Mode→Track. Now push the soft key to the right of CurA. Note the similar circle-arrow that is lit up on the right side of the o-scope. Turn this knob and the cursor should track the waveform. The values of voltage and time should be displayed on the screen. Use the cursor to find the peak and trough values and the times at which they occur. This is the data that fills in Table 1.